AquaVision AI – UI Wireframes

Low-fidelity screens for the React + Vite frontend · reference for building the Figma file

Screens in this file

0 Who's working? – start screen name picker
1 Upload (home) – drag and drop, plus 'What's left' work queue
1b Upload (home) – dark mode, theme switch
2 Batch processing – progress, tab title, finished alert
3 Results and review – view-only zoom/brightness, 15 points
3b Keyboard shortcuts (press ?)
3c Export check before export
4 Results – image batch grid
5 Label – unknowns gallery, frame canvas, label picker
6 Add new species – confirmation dialog
7 QC – blind count
8 Trends – clarity index and species found per video
9 Species registry
10 Settings, benchmark and storage

Conventions



Grey box with a cross = video frame or image



Teal box = AI detection; amber = needs attention

✓ Done ⚠ Needs review × Failed – Skipped

Status chips always use icon + text, never colour alone

1 Red numbers point to the notes under each screen

Primary Secondary Disabled

Buttons: primary (one per screen), secondary, disabled; all with keyboard focus style

Figma tips

- Frame size 1440 × 900 (desktop only); check at 1280 × 800 for laptops.
- Name frames after React components (UploadDropzone, LabelPicker, TrendsChart…).
- Build components with variants: status chip, button, dropzone, unknown card, point marker.
- Use Auto Layout and an 8 px grid; font Inter (bundled locally in the app, never from Google Fonts).
- Link frames into a prototype following Demo A then Demo B; it doubles as a backup demo.

0 · Who's working? (start screen)

Shown when the app starts · no passwords · the name is saved with every edit, label and QC count

AquaVision AI | Batch: Fall 2025 · August 20 · T1 ▾ | 4 ◉ Alex (analyst) ▾ | ? Shortcuts | ☀ Light ☾ Dark | ● Offline | GPU mode

- ⬆ Upload
- ▤ Results
- ✎ Label 12
- ✓ QC
- ↗ Trends
- ◎ Species
- ⚙ Settings

Upload survey media

⬆

Drag videos, images, ZIP files or folders here

Batch: 5 items · 7.9 GB Disk: 112 GB free

File (path in upload)
T1/STAT 1/GX010021.MP4 + GX020021.MP4 (joined)
T1/STAT 2/GX010022.MP4
photos.zip → 12 images
__MACOSX/._GX010021
T2/notes.docx

What's left

click a number to go straight to that ta

Batch	Wind				Failed files	
Fall 2025 · Aug 20 · T1	1 →				1 →	Continue
Spring 2026 · May 14 · T2	0				0	Continue
photos.zip (images)	—	1 →	2 →	—	0	Continue

☐ Use files in place (read from external drive, no copy)

Clear batch | Process batch →

Who's working today?

Your name is saved with your edits, labels and QC counts.

1

- A **Alex** — analyst
- J **Jordan** — analyst
- P **Priya** — QC reviewer
- S **Sam** — monitoring lead

+ Add a name 2

Blind QC will ask for a different name than the first counter. 3

Continue →

1 Pick your name once per session; no password (single local computer)
2 Add a name the first time someone uses the app
3 Blind QC needs two different names; labels and edits record who made them
4 Switch user anytime from the name in the top bar

1 · Upload (home)

UploadDropzone + UploadProgress · one upload = one batch · the app opens on this page

AquaVision AI | Batch: Fall 2025 · August 20 · T1 ▾ | ◉ Alex (analyst) ▾ | ? Shortcuts 7 | ☀ Light | ☾ Dark | ● Offline | GPU mode

- ⬆ Upload
- ▤ Results
- ✎ Label 12
- ✓ QC
- ↗ Trends
- ◎ Species
- ⚙ Settings

Upload survey media

1

⬆

Drag videos, images, ZIP files or folders here

Choose files

Video .mp4 .mov .avi .mkv · Image .jpg .jpeg .png .tif .tiff · Archive .zip 2

Batch: 5 items · 7.9 GB Disk: 112 GB free

File (path in upload)	Type	Size	Progress	Status
3 T1/STAT 1/GX010021.MP4 + GX020021.MP4 (joined)	Video	3.8 GB	4 64%	↑ Uploading
T1/STAT 2/GX010022.MP4	Video	3.9 GB	100%	✓ Done
photos.zip → 12 images	Archive	48 MB	100%	✓ Done
__MACOSX/._GX010021	—	4 KB	—	– Skipped: system file
T2/notes.docx	—	22 KB	—	– Skipped: unsupported type

5

What's left

click a number to go straight to that task 8

Batch	Windows to confirm	Points to review	Unknowns to label	QC pending	Failed files	
Fall 2025 · Aug 20 · T1	1 →	2 →	12 →	4 of 6 →	1 →	Continue
Spring 2026 · May 14 · T2	0	0	3 →	✓	0	Continue
photos.zip (images)	—	1 →	2 →	—	0	Continue

☐ Use files in place (read from external drive, no copy) 6

Clear batch | Process batch →

1 Drop area accepts files, multiple files and folders; 'Choose files' for keyboard users
2 Accepted types come from app.yaml
3 Folder paths kept: Transect / STAT / Video ID read from them
4 Large videos upload in chunks: % per file and for the batch; failed chunks retry
5 Skipped files show a reason and never fail the batch
6 Option to read files in place from the external drive (P1)
7 Light / Dark switch in the top bar (see 1b)
8 What's left: per-batch work queue; each number opens that task, so staff can resume anytime

1b · Upload (home) – dark mode

Same screen with the theme switch set to Dark · the choice is remembered until changed

AquaVision AI | Batch: Fall 2025 · August 20 · T1 ▾ | ◉ Alex (analyst) ▾ | ? Shortcuts 1 | ☀ Light | ☾ Dark | ● Offline | GPU mode

- ⬆ Upload
- ▤ Results
- ✎ Label 12
- ✓ QC
- ↗ Trends
- ◎ Species
- ⚙ Settings

Upload survey media

⬆

Drag videos, images, ZIP files or folders here

Choose files

Video .mp4 .mov .avi .mkv · Image .jpg .jpeg .png .tif .tiff · Archive .zip

Batch: 5 items · 7.9 GB Disk: 112 GB free

File (path in upload)	Type	Size	Progress	Status
T1/STAT 1/GX010021.MP4 + GX020021.MP4 (joined)	Video	3.8 GB	64%	↑ Uploading
T1/STAT 2/GX010022.MP4	Video	3.9 GB	100%	✓ Done
photos.zip → 12 images	Archive	48 MB	100%	✓ Done
__MACOSX/._GX010021	—	4 KB	—	– Skipped: system file
T2/notes.docx	—	22 KB	—	– Skipped: unsupported type

What's left

click a number to go straight to that task

Batch	Windows to confirm	Points to review	Unknowns to label	QC pending	Failed files	
Fall 2025 · Aug 20 · T1	1 →	2 →	12 →	4 of 6 →	1 →	Continue
Spring 2026 · May 14 · T2	0	0	3 →	✓	0	Continue
photos.zip (images)	—	1 →	2 →	—	0	Continue

☐ Use files in place (read from external drive, no copy)

Clear batch | Process batch →

1. Light / Dark switch in the top bar on every page, including this home page
2. Choice saved on this computer and restored on the next start, until the user changes it
3. First start (nothing saved yet) follows the computer's light/dark setting
4. Switching is instant; no reload, no processing interrupted
5. Dark palette checked for contrast; status chips keep icon + text
6. Detection boxes, charts and video frames use dark-mode colours too

2 · Batch processing

Background job · frontend polls job status

(58%) AquaVision AI – Processing × 7

AquaVision AI | Batch: Fall 2025 · August 20 · T1 ▾ | ◉ Alex (analyst) ▾ | ? Shortcuts | ☀ Light / ☾ Dark | ● Offline | GPU mode

- ⬆ Upload
- ▤ **Results**
- ✎ Label 12
- ✓ QC
- ↗ Trends
- ◎ Species
- ⚙ Settings

Processing batch – Fall 2025 · August 20 · T1

1

58% · Step: species identification · about 6 min left (GPU)

☑ Notify me when done (desktop notification)

Video	Pipeline (Ingest · Window · Points · Detect · Species · Count · Clarity) 2	Analysis window 3	Status	6
T1 · STAT 1 · GX010021	● ● ● ● ● ● ●	01:00 – 16:00 ✓ confirmed	✓ Done	**Open**
T1 · STAT 2 · GX010022	● ● ● ● ● ○ ○	03:10 – 18:10 (sediment)	↻ Processing	
T1 · STAT 3 · GX010023	● ● ○ ○ ○ ○ ○	Proposed 01:00 – 16:00	⚠ Confirm window 4	**Confirm**
T1 · STAT 4 · GX010024	○ ○ ○ ○ ○ ○ ○	—	× Unreadable file	**Details**
12 images (photos.zip)	○ ○ ○ ○ ○ ○ ○	n/a (images)	○ Waiting	

Mode: GPU (12 GB) · peak VRAM 3.9 GB 5

Average 4 min 10 s per 20-min video (measured)

✓ Batch finished · Fall 2025 · Aug 20 · T1 8

4 videos ready · 1 failed (T1 · STAT 4) · 12 unknowns to label

Open results | **Dismiss**

1 Overall progress and current step for the batch
2 Per-video pipeline steps so staff see where time is spent
3 Analysis window proposed per video; staff confirm before counting
4 Errors are per file; the rest of the batch continues
5 Mode and time per video feed the benchmark
6 Results open per video when ready
7 Browser tab shows progress, so staff can work in other windows
8 When finished: in-app message and optional desktop notification

3 · Results and review – one video

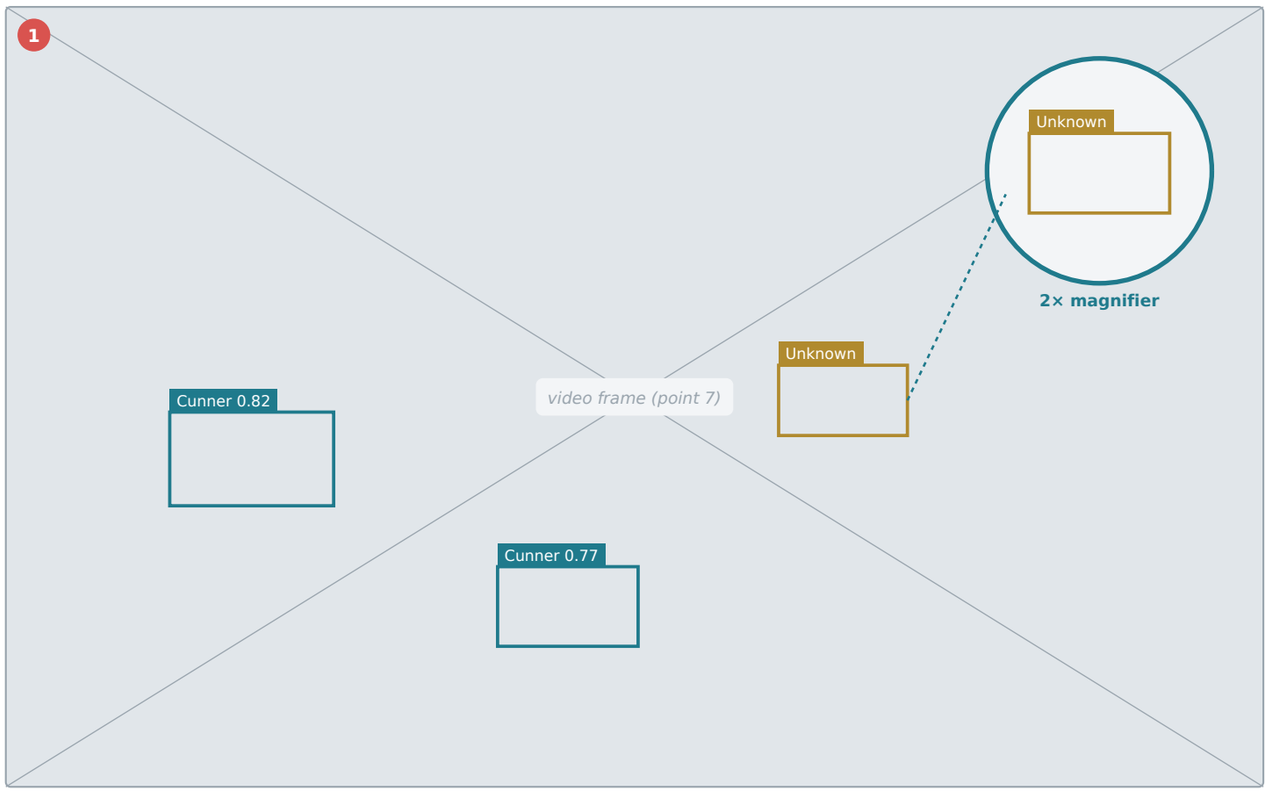
VideoPlayer (canvas overlay) + point timeline + point panel

AquaVision AI | Batch: Fall 2025 · August 20 · T1 ▾ | ◉ Alex (analyst) ▾ (8) | ? Shortcuts | ☀ Light ☾ Dark | ● Offline | GPU mode

- ⬆ Upload
- ▤ **Results**
- ✎ Label 12
- ✓ QC
- ↗ Trends
- ◎ Species
- ⚙ Settings

T1 · STAT 1 · GX010021 · Aug 20 2025 · Window 01:00 – 16:00

View only: Zoom − 2× + | Brightness | Contrast | **Reset** | ↺ Loop ±0.5 s ON (7)



▶ ❚❚ 06:00.0 / 20:12 | ◀ 1 frame ▶ | ±0.5 s window highlighted | 0.25× 0.5× 1×

Subsample points

1 2 3 4 5 6 **7** 8 9 10 11 12 13 14 15 (2)

✓ done ⚠ needs review ◆ QC sample ■ current

Point 7 · 06:00 (3)

Total fish	**3**
Invertebrates	1 (crab)
Cunner	2
Unknown / unable to ID	1 → Label
Species observed	1
Clarity index	62 (estimate)
Visibility	*Suggested: 0.5–1.0 m*

(5) Confirm: 0.5–1.0 m ▾

Notes

Fish behind kelp at 06:03 not counted (outside ±0.5 s)

Save edits | **Undo**

(4) **MeanCount 2.4 · Species found 3 · Clarity index 58 (worsening) · ⚠ 2 points need review · 1 unknown left**

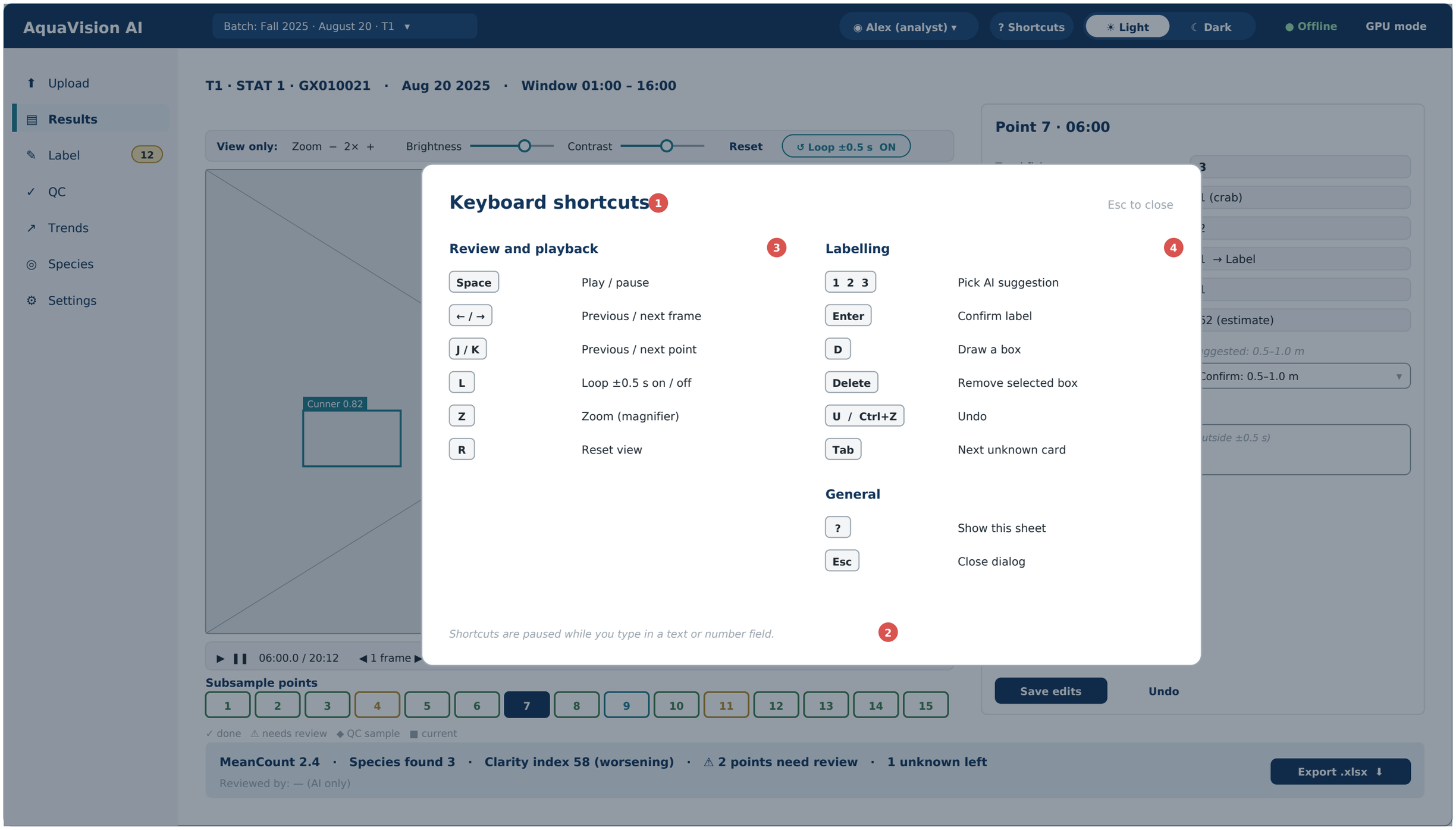
Reviewed by: — (AI only)

(6) **Export .xlsx ⬇**

1. Video with AI boxes drawn on a canvas layer; never re-encoded
2. 15 subsample points; colour + icon show status; click to jump
3. Point panel: counts, species, Unknown, clarity, visibility; editable
4. Video summary: MeanCount, species found, clarity, items needing review
5. Confirmed visibility goes to the export; suggested value stays separate
6. Export always available; runs the export check (3c)
7. View-only zoom, brightness/contrast and ±0.5 s loop; never change the analysis
8. '? Shortcuts' in the top bar opens the keyboard sheet (3b)

3b · Keyboard shortcuts (press ?)

Overlay available from every screen · every shortcut also has a visible button



1 Opens with ? or the 'Shortcuts' button in the top bar; Esc closes
2 Shortcuts never fire while typing in a text field or number box
3 Same keys on every screen where they apply
4 Optional: every action also has a visible button

3c · Export check

Shown when Export is clicked · warns about open items, never blocks

AquaVision AI | Batch: Fall 2025 · August 20 · T1 | Alex (analyst) | ? Shortcuts | Light | Dark | Offline | GPU mode

Upload · Results · Label 12 · QC · Trends · Species · Settings

T1 · STAT 1 · GX010021 · Aug 20 2025 · Window 01:00 – 16:00

View only: Zoom − 2× + | Brightness | Contrast | Reset | Loop ±0.5 s ON

Cunner 0.82

▶ ❚❚ 06:00.0 / 20:12 ◀ 1 frame ▶ ±0.5 s window highlighted 0.25× 0.5× 1×

Subsampling points

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

✓ done ⚠ needs review ◆ QC sample ■ current

MeanCount 2.4 · Species found 3 · Clarity index 58 (worsening) · ⚠ 2 points need review · 1 unknown left

Reviewed by: — (AI only)

Export .xlsx ⬇

Point 7 · 06:00

Total fish 3

Invertebrates 1 (crab)

2

1 → Label

1

62 (estimate)

Suggested: 0.5–1.0 m

Confirm: 0.5–1.0 m

not counted (outside ±0.5 s)

Save edits | Undo

Before you export · T1 · STAT 1 (1)

- ✓ Analysis window confirmed (2)
- ⚠ 1 unknown fish not labelled — Label →
- ⚠ 2 points need review — Review →
- ⚠ Visibility not confirmed on 3 points — Review →
- ⚠ QC agreement 80% (target 95%) — QC →
- ✓ All processing steps finished

(4) *Exporting now marks the file 'Draft' and lists the 4 open items on a 'Read me' sheet.*

Go to first issue | (3) Export anyway | Cancel

1. Runs every time Export is clicked
2. Each open item links to the screen that fixes it
3. Never blocks: 'Export anyway' is always available
4. Exports with open items are marked 'Draft' and list the items on a 'Read me' sheet

4 · Results – image batch

Images get per-image counts; never part of MeanCount; exported on the 'Images' sheet

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- ⬆ Upload
- ▤ **Results**
- ✎ Label 12
- ✓ QC
- ↗ Trends
- ◎ Species
- ⚙ Settings

photos.zip · 12 images

3 Folder: all ▾ | Status: all ▾

1 Cunner 0.8 — image 01.jpg	fish — image 02.jpg (4)	image 03.jpg	fish — image 04.jpg
Fish 1 · Invert. 0 · Unknown 0	Fish 2 · Invert. 1 · Unknown 0	Fish 3 · Invert. 0 · Unknown 1	Fish 1 · Invert. 1 · Unknown 0
✓ Reviewed	◇ AI only	⚠ Needs review	✓ Reviewed
2			
Cunner 0.8 — image 05.jpg	image 06.jpg	Cunner 0.8 — image 07.jpg	fish — image 08.jpg
Fish 2 · Invert. 0 · Unknown 0	Fish 3 · Invert. 1 · Unknown 1	Fish 1 · Invert. 0 · Unknown 0	Fish 2 · Invert. 1 · Unknown 0
◇ AI only	⚠ Needs review	✓ Reviewed	◇ AI only

1 Grid of images with boxes; counts under each
2 Status per image: reviewed, needs review, AI only
3 Filters by folder and status
4 Opening an image goes to the same review panel (no timeline)

5 · Label unknowns

UnknownGallery + LabelDialog (canvas) + LabelPicker · the most important screen to get right

AquaVision AI | Batch: Fall 2025 · August 20 · T1 ▾ | ◉ Alex (analyst) ▾ | ? Shortcuts | ☀ Light / ☾ Dark | ● Offline | GPU mode

- ⬆ Upload
- ▤ Results
- ✎ **Label** 12
- ✓ QC
- ↗ Trends
- ◎ Species
- ⚙ Settings

Unknowns · 12 left

Video ▾ | T1 ▾ | Unlabelled ▾

1

crop
T1 · P7 · 06:00
Low confidence
☑ Select

crop
T1 · P9 · 08:00
Novelty: far from all
☑ Select

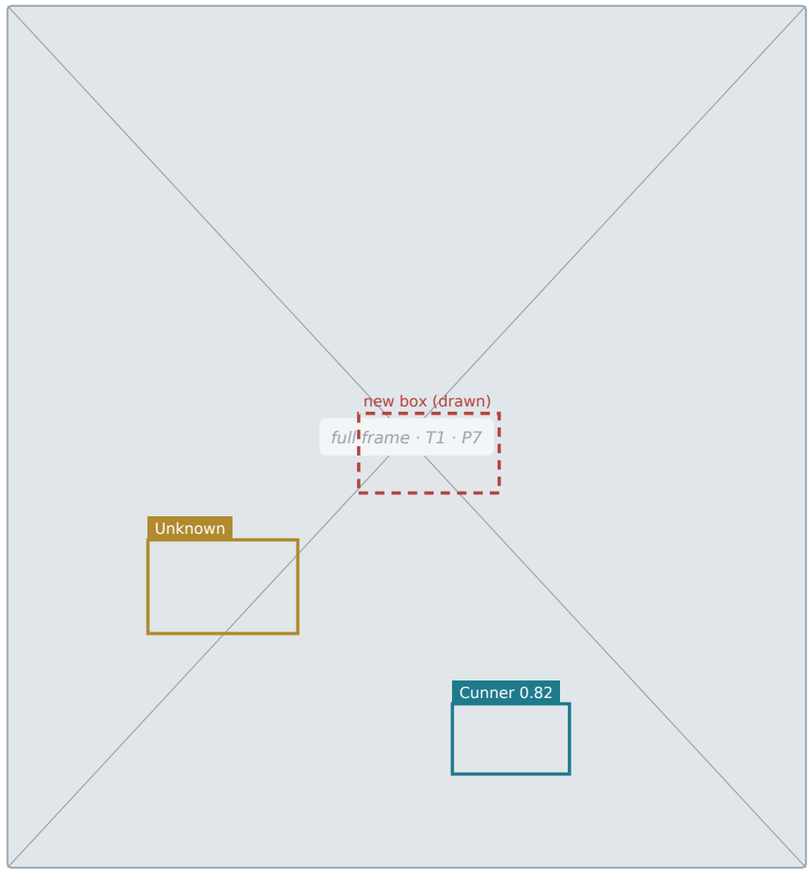
crop
T2 · P3 · 03:00
Low confidence
☐ Select

crop
T2 · P11 · 10:00
Low visibility
☐ Select

crop
IMG 07
Low confidence
☐ Select

crop
T3 · P2 · 02:00
Novelty
☐ Select

2 selected



+ Draw box | Adjust | Delete box | ◀ Frame | Frame ▶ 2

Frame 10 803 · 06:00.2 · inside ±0.5 s of point 7 → counts

Label 2 selected

AI suggestions 3

1 Cunner 0.62
2 Pollock 0.21
3 W. flounder 0.09

Or choose

Choose a label… ▾ 4

Fish species
★ Cunner (Tautogolabrus adspersus)
Pollock (Pollachius virens)
Invertebrates
crab shrimp worm snail
Other
Unable to ID due to visibility
Not a fish / not an organism

+ Add new species… 5

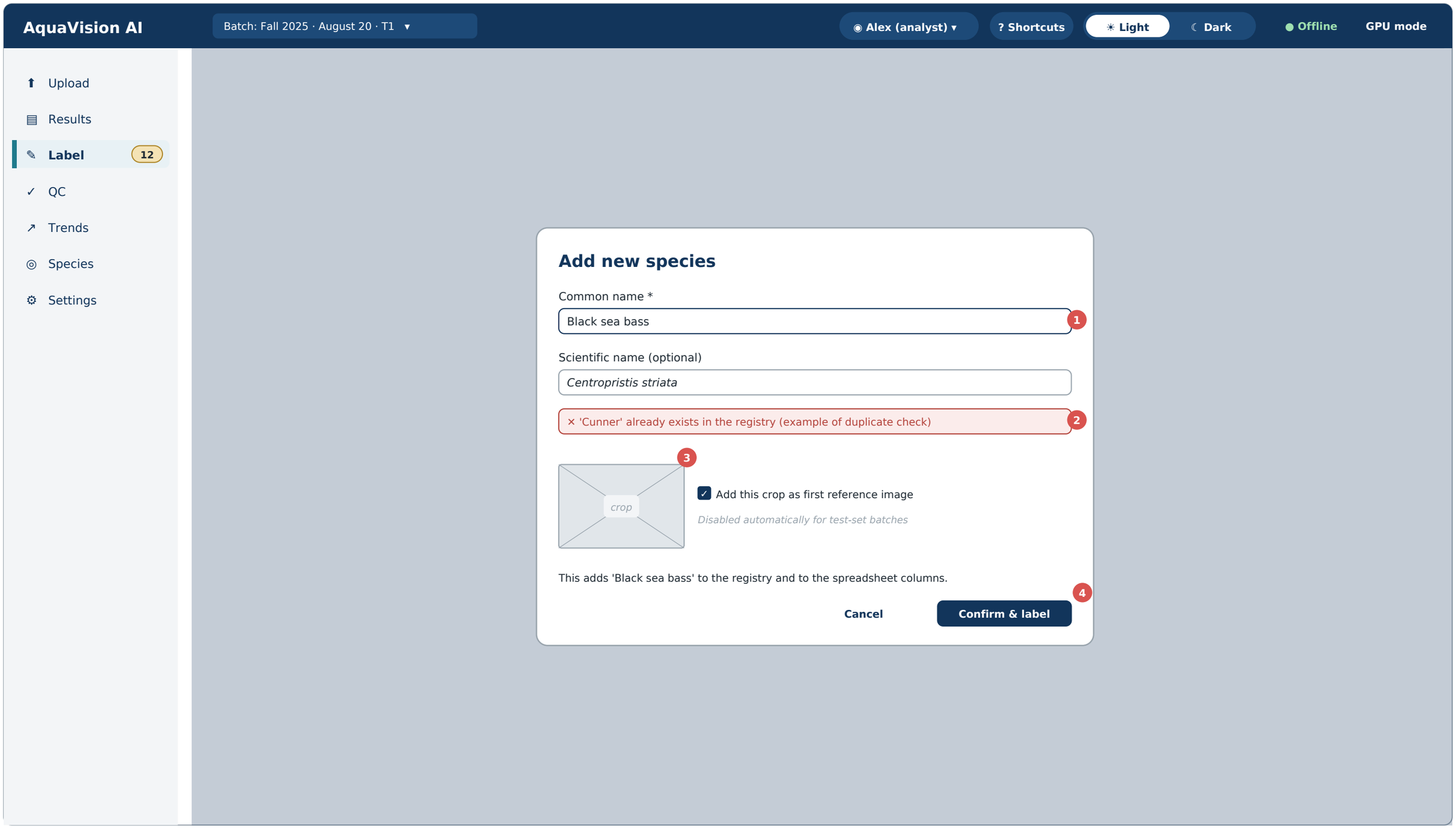
☐ Use as reference image 6

Apply to 2 selected | ↶ Undo

1 One card per tracked organism (not per frame); multi-select for bulk labelling
2 Canvas: draw new box, resize/move, delete; step frames to spot camouflaged fish
3 AI top 3 suggestions as buttons; keys 1–3 then Enter
4 Grouped dropdown: Fish species / Invertebrates / Other; no free text
5 'Add new species…' opens a dialog with confirmation (screen 6)
6 Reference image option disabled for test-set batches; every action can be undone

6 · Add new species (dialog)

NewSpeciesForm · opened from the label picker; nothing is created without confirmation



1 Common name required; scientific name optional
2 Duplicate names (ignoring case and spaces) are blocked
3 The fish crop can become the first reference image
4 Two-step: review, then confirm

7 · QC – blind count

Random ≥10% of points per video · AI and other counts hidden until submitted

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- ⬆ Upload
- ▤ Results
- ✎ Label 12
- ✓ **QC**
- ↗ Trends
- ◎ Species
- ⚙ Settings

Blind QC · T1 · STAT 1 · point 4 of 6 (random 10%)

video at point · AI boxes hidden

► ❚❚ ◀ 1 frame ▶ ±5 s context 0.25× 0.5× 1×

Your count

Fish	2	− +
Invertebrates	0	− +

AI count: hidden · Other reviewer: hidden (2)

Submit count

After submit (previous point) (3)

You: 2 · AI: 3 · Difference 1 fish (33%)

⚠ Above 6% – recount by another person

Video agreement so far: 4 / 5 points (80%) · target 95% (4)

1. Progress through the random QC sample
2. AI and other reviewers' counts are hidden
3. After submit: percent and absolute difference; >6% flagged for recount
4. Agreement per video vs. the 95% target

8 · Trends

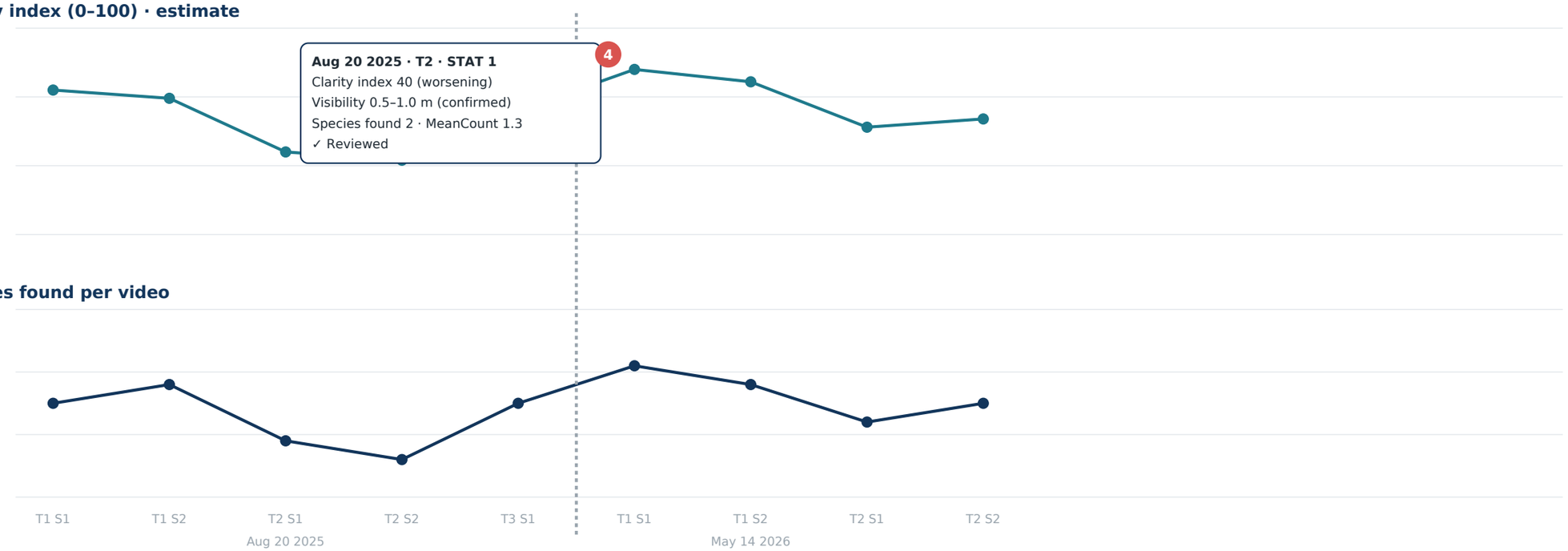
TrendsChart · one point per video in recording-date order · clarity index and species found stacked

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- ⬆ Upload
- ▤ Results
- ✎ Label 12
- ✓ QC
- ↗ **Trends**
- ◎ Species
- ⚙ Settings

Transect: T1 ▾ | STAT: all ▾ | Aug 2025 – Nov 2026 ▾ (1) | ◉ Reviewed only | ○ Include AI-only | ☐ Group by survey (2)

Clarity index (0–100) · estimate



2 videos without a recording date (not plotted) · fix dates → (5)

Clarity index is an image-based estimate, not turbidity (NTU) (6)

1. Filters: Transect, STAT, date range
2. Reviewed-only by default; toggle to include AI-only; group by survey
3. Two charts share one x-axis so species drops can be checked against clarity
4. Tooltip: date, Transect, STAT, clarity, visibility, species found, MeanCount
5. Videos without a date are listed separately, not plotted
6. Clarity index is an estimate (not turbidity)

9 · Species registry

Species list used by the label picker, species ID and export columns

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- ⬆ Upload
- ▤ Results
- ✎ Label 12
- ✓ QC
- ↗ Trends
- ◎ **Species**
- ⚙ Settings

Species registry · 6 species

1 Search species... | + Add species

Common name	Scientific name	Reference images 2	Added by 3	Export column 4
Cunner	Tautogolabrus adspersus	☒ ☒ ☒ ☒ ☒	Registry seed	✓
Pollock	Pollachius virens	☒ ☒ ☒	Registry seed	✓
Winter flounder	Pseudopleuronectes americanus	☒ ☒ ☒ ☒	Registry seed	✓
Atlantic tomcod	Microgadus tomcod	☒ ☒	Registry seed	✓
Rock gunnel	Pholis gunnellus	☒	Analyst · Nov 6 2026	✓
Black sea bass	Centropristis striata	☒	Analyst · Nov 6 2026	✓ new

Species names shown are examples; the real list comes from Coastal Action.

1 Search and add species
2 Reference images per species; remove if wrong
3 Who added a species and when
4 New species become export columns

10 · Settings and benchmark

Protocol values from protocol.yaml (draft is still changing) · hardware benchmark

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- ⬆ Upload
- ▤ Results
- ✎ Label 12
- ✓ QC
- ↗ Trends
- ◎ Species
- ⚙ **Settings**

Protocol settings

Trim start / end (s)	60 / 60 (1)
Analysis window (min)	15
Subsample interval (s)	60 → 15 points
Context around point (s)	±5
Count tolerance (s)	±0.5 (max 1.0)
QC sample	10%
QC max difference	6%
Detection confidence	0.45
Low-clarity review threshold	35

Save | **Reset to defaults**

Applies to new processing only (2)

Benchmark

Metric	GPU (12 GB)	CPU laptop
Time per 20-min video	measured	measured
Frames processed / s	measured	measured
Peak VRAM	measured	—
Peak RAM	measured	measured
Clarity index overhead	measured	measured

(3)

Storage

Data folder: D:\AquaVision\data

(5)

112 GB free of 1 TB · warning below 50 GB

What uses space	Size
Copied videos (uploads)	46.2 GB
Context clips and frames	8.4 GB
Database and exports	0.3 GB

Remove copied videos of exported batches… (6)

Asks first · originals on the drive are never touched

Run benchmark | **Export for hardware report** (4)

1. Protocol settings editable by an admin; defaults from the current draft
2. Changes apply to new processing only; old results keep their settings
3. GPU/CPU mode and timings measured, never estimated
4. Benchmark exportable for the hardware report
5. Storage: free space and what uses it; warning in the top bar when space runs low
6. Remove copied videos of exported batches (asks first); originals on the drive are never touched